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Course Code	ITCD 103	Section	29104
Course Details	Software Development for Enterprise Systems	Date	02/26/26
Names	Allan Mark Gardoce	Hands-On Laboratory	

Chapter 3: Software Development Methodology
Hands-on Laboratory Exercise 3

Objective: This laboratory exercise aims to enhance your understanding of HRMS systems and their components by designing a detailed diagram using Draw.io. Through this exercise, you will:

1. Learn the waterfall software development lifecycle (SDLC) through a practical HRMS project.
2. Gain experience using Draw.io for system design and documentation.
3. Understand the key components and functionalities of an HRMS.
4. Apply best practices for system design and user interface (UI) flow.

Lab Environment:

- Computer with internet access
- Draw.io

Steps:

1. Requirements:

- Use Draw.io to create a flowchart representing the overall system flow.
- Identify key actors (e.g., HR personnel, employees) and their interactions with the system.
- Define functional requirements for each module (e.g., adding new employees, generating paychecks).
- Use text boxes and shapes to represent requirements and their relationships.

2. Design:

- Design the user interface (UI) for each module using shapes and connectors.
- Define data structures and workflows for each functionality.
- Use mockups and diagrams to visualize the design.

3. Development:

- Simulate the development process by creating mock data and workflows in Draw.io.

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- Show how data flows through different modules and how functionalities are implemented.
- You can use different colors and shapes to represent different data types and processes.

4. Testing:

- Design test cases for each functionality and document them in Draw.io.
- Use conditional shapes and text boxes to represent different test scenarios and expected outcomes.

5. Deployment:

- Create a deployment plan outlining how the system will be rolled out to users.
- Use a flowchart or timeline diagram to show the steps involved.

6. Maintenance:

- Show how the system can be updated and maintained in Draw.io.
- Use diagrams to represent bug fixes, new features, and system upgrades.

Conclusion:

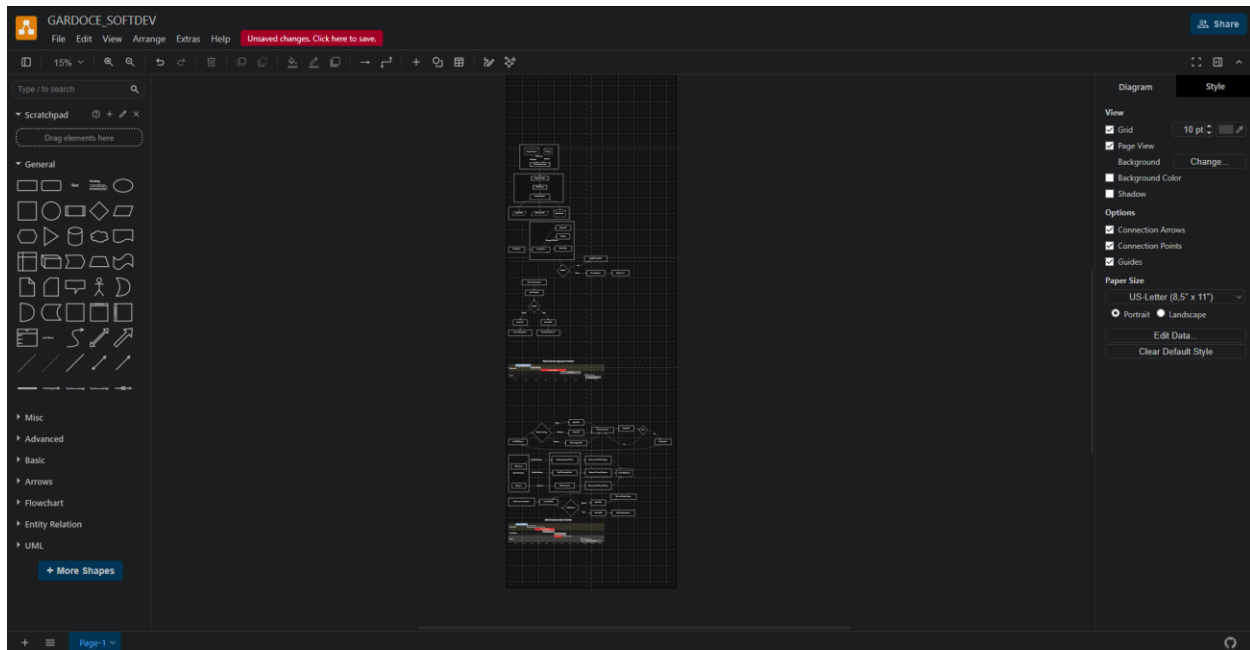
The execution of this laboratory exercise underscores the vital role of the Waterfall methodology in establishing a disciplined framework for enterprise-level system development. By systematically navigating through the initial requirements and design phases, I was able to transform abstract organizational needs into a concrete, logical structure for a Human Resource Management System. This sequential progression ensures that every functional requirement, such as payroll processing and employee data management, is grounded in a clear architectural vision before technical simulation begins. Such a structured approach is indispensable for minimizing risks and ensuring that the final system aligns perfectly with the operational demands of both administrative staff and general employees.

Beyond the theoretical application of the SDLC, the practical use of Draw.io served as a bridge between conceptual documentation and visual system architecture. Mapping out the complexities of data flow, testing scenarios, and maintenance cycles provided a deeper appreciation for the long-term viability of software. This exercise demonstrated that a successful HRMS is not merely about functional code, but about a holistic lifecycle that accounts for rigorous quality assurance and the evolving nature of post-deployment support. Ultimately, gaining proficiency in these visualization techniques equips a developer with the clarity needed to manage large-scale enterprise projects with both technical precision and professional foresight.

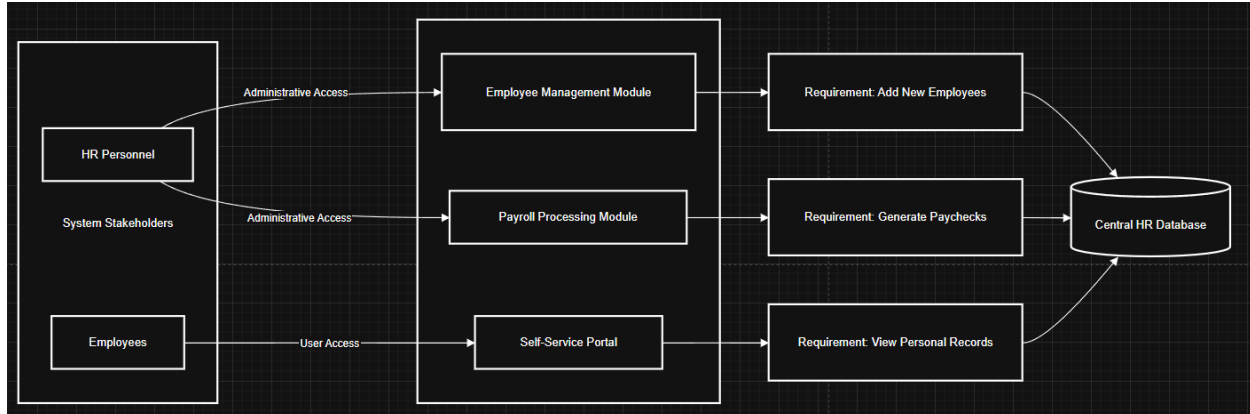
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Print Screen the Visual Representaion:

Draw.io Whole Screen:

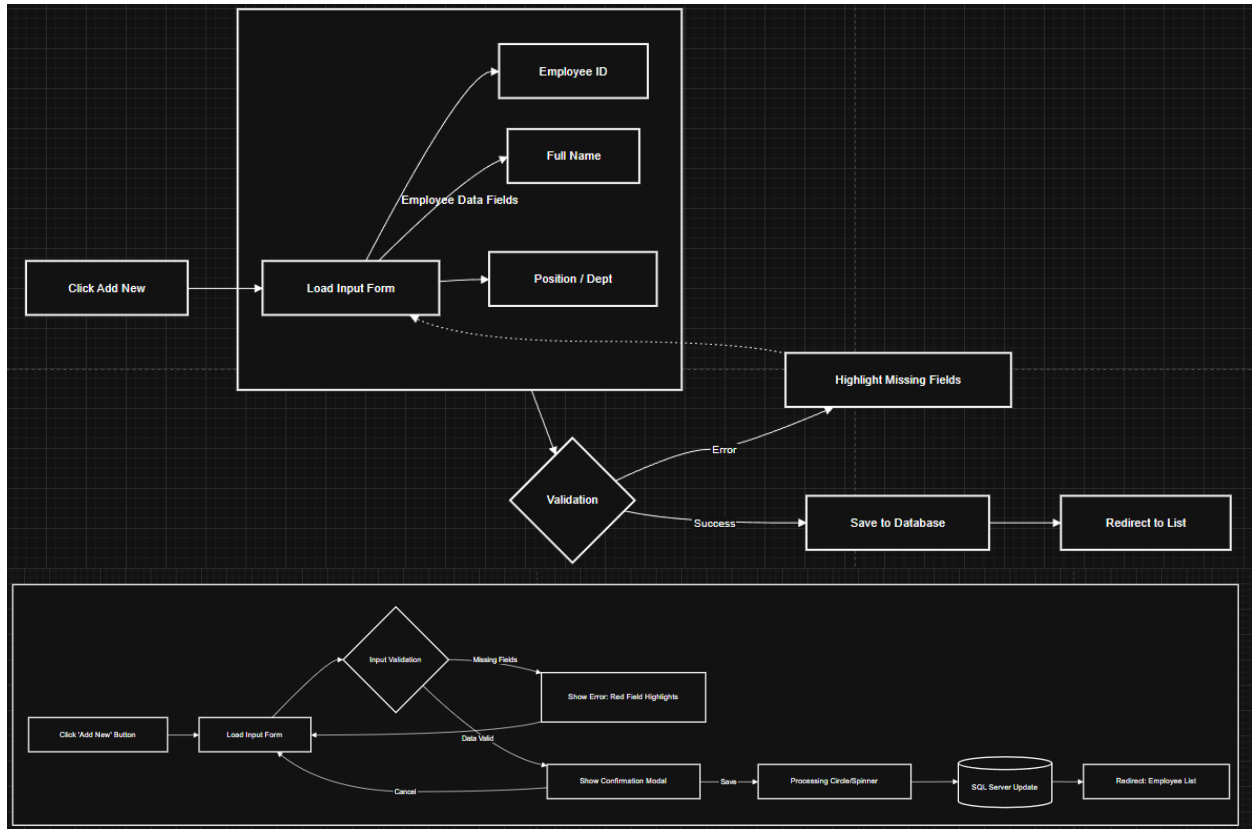


STEP 1:



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STEP 2:



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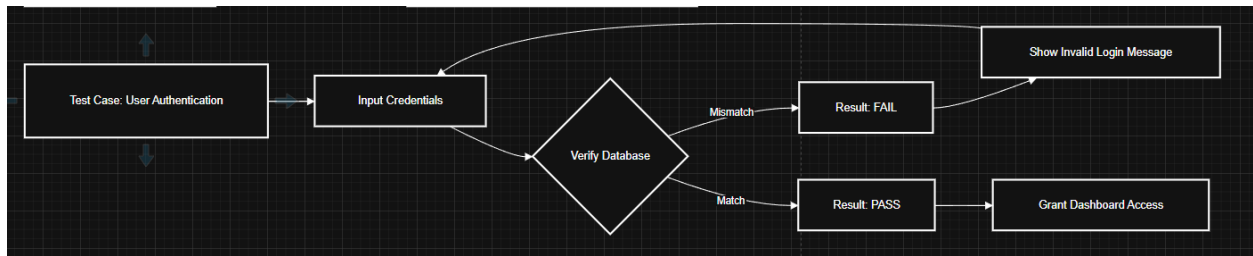
STEP 3:



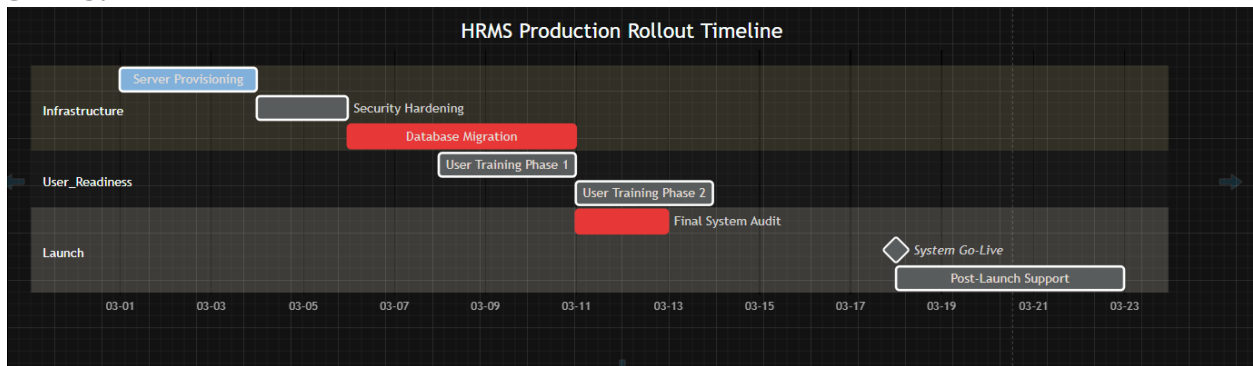
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STEP 4:



STEP 5:



STEP 6:

